

KIRUTHIKA G (192312306)

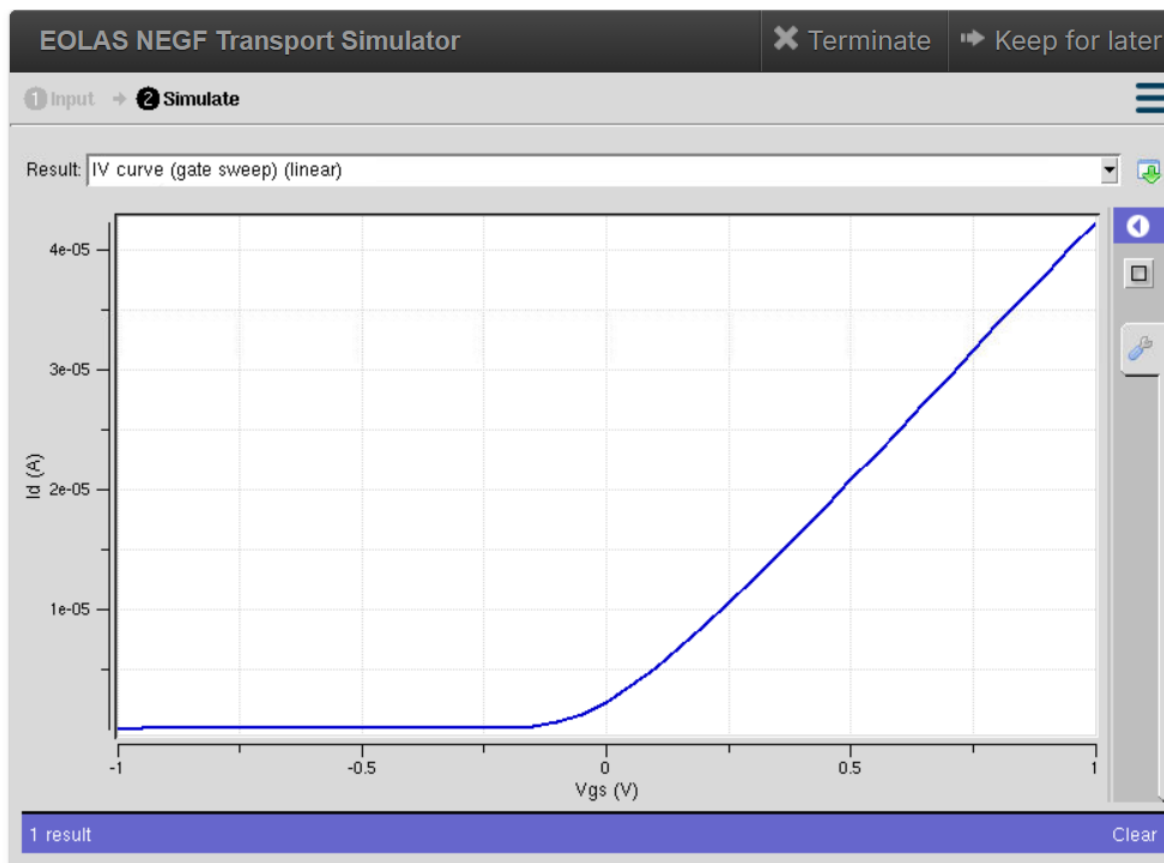
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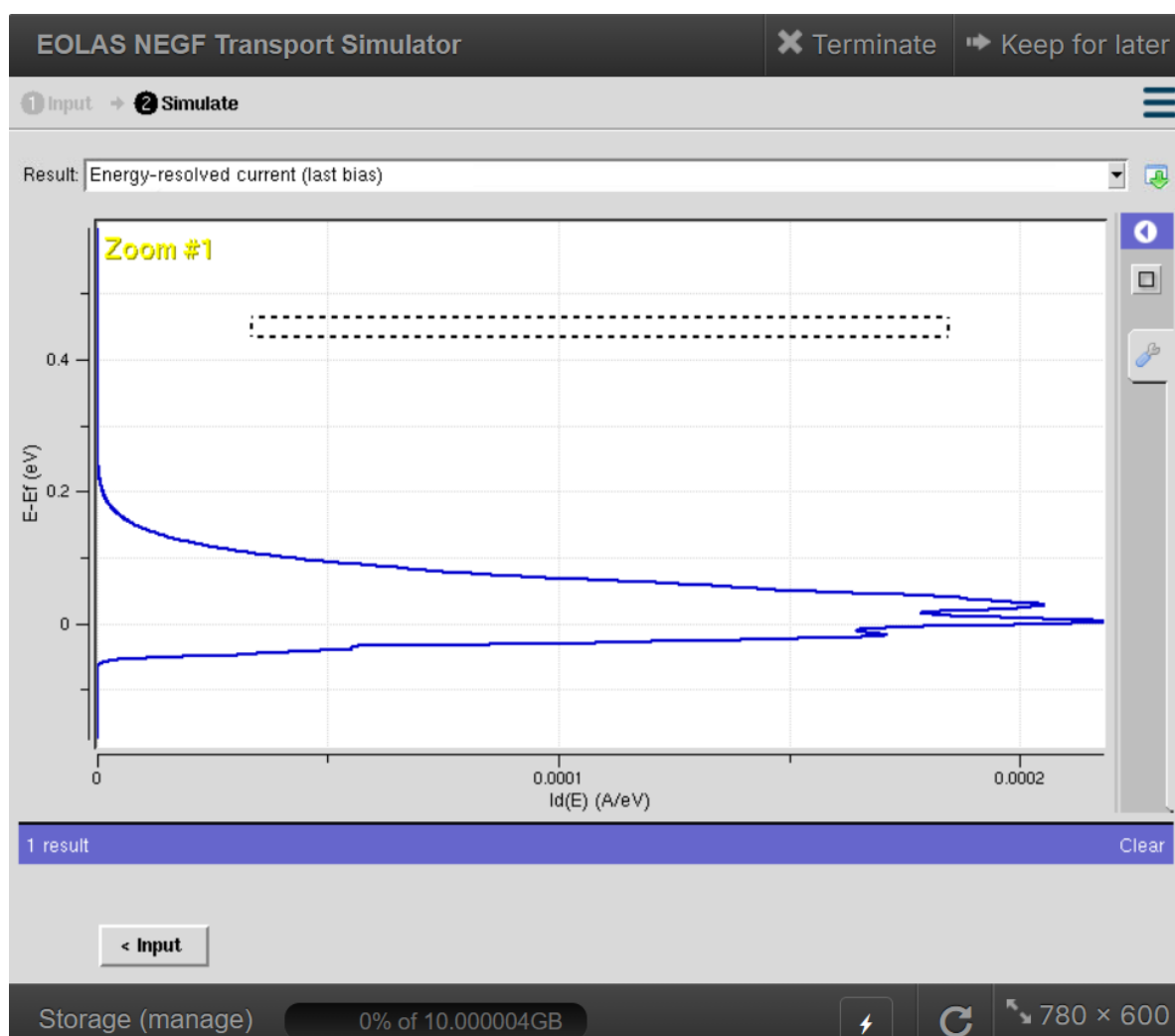
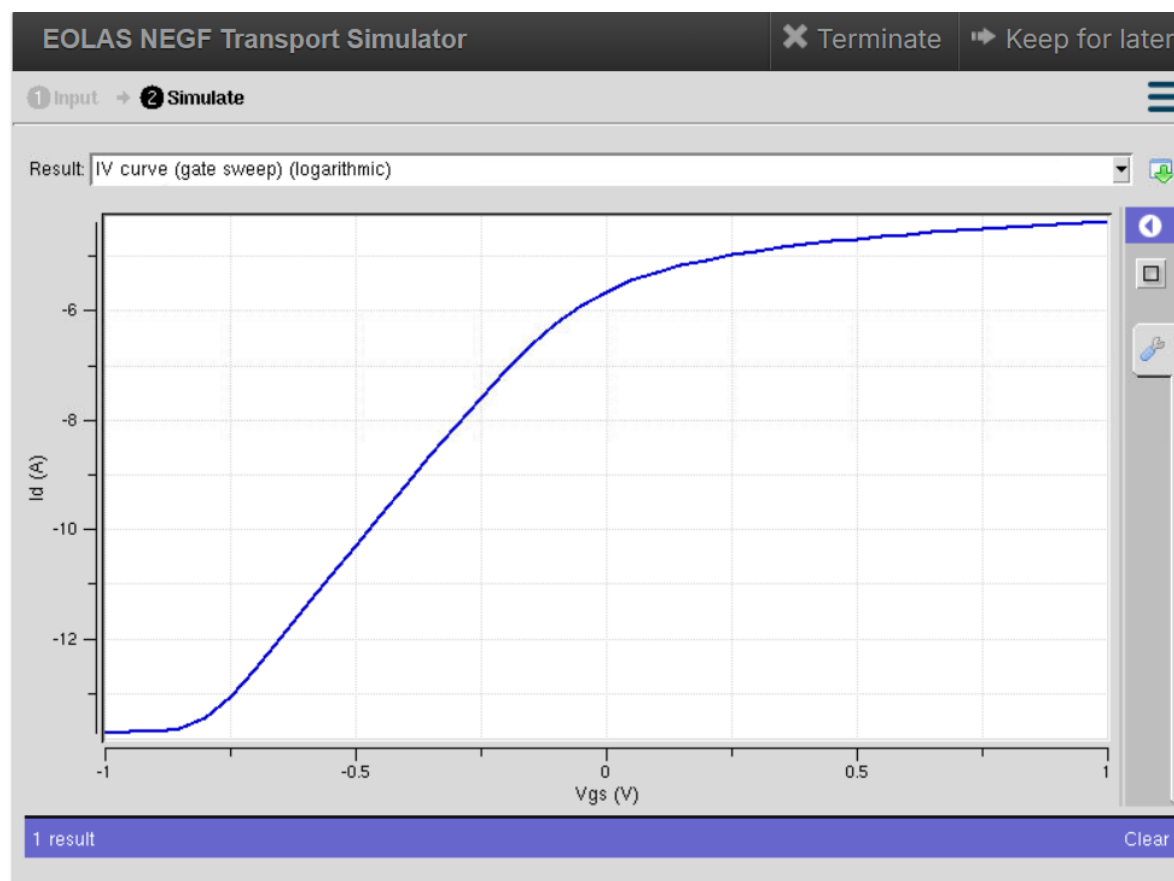
NANO ELECTRONICS

Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET
using NanoHub FINFET

Output for finFET

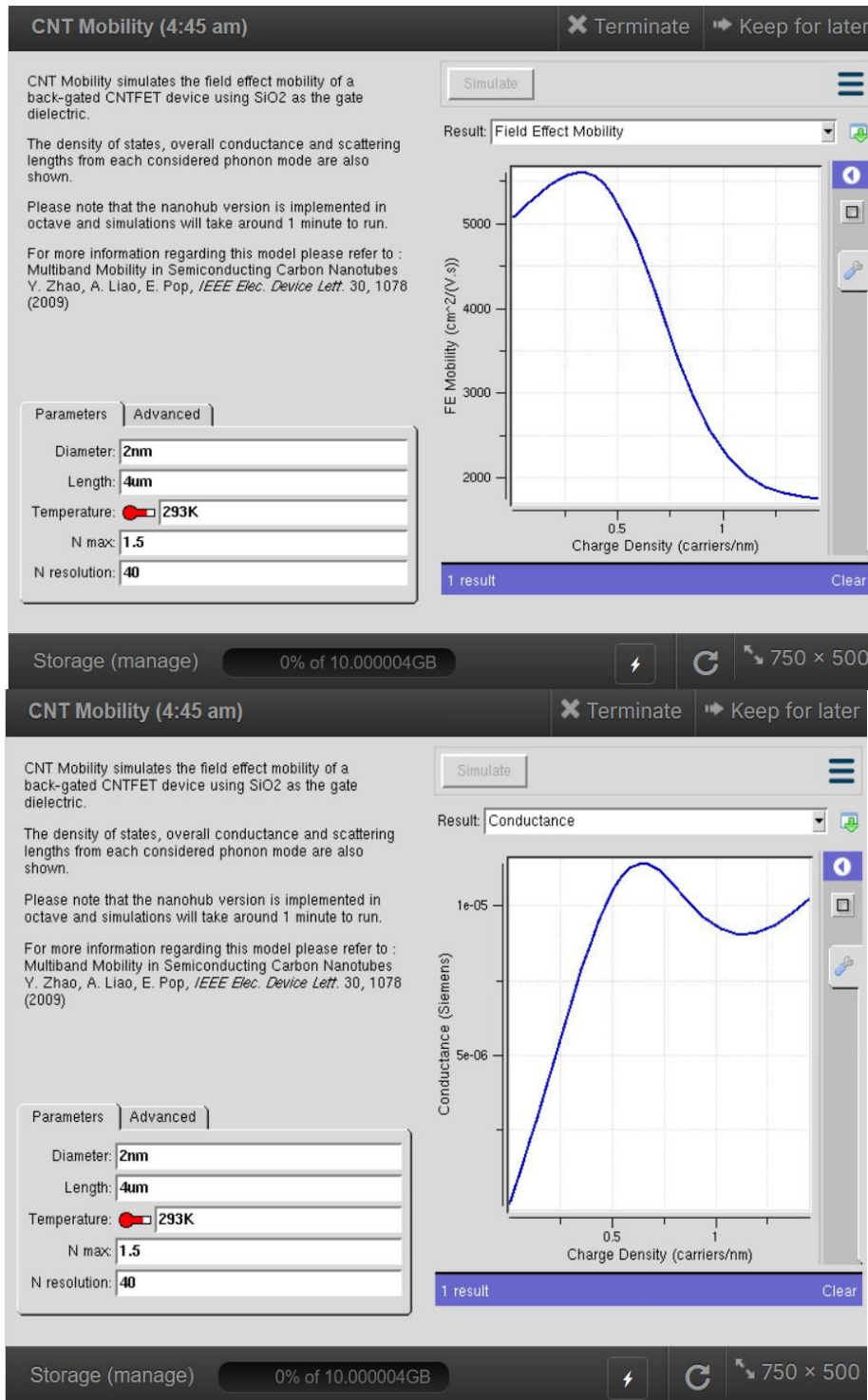
FinFET stands for Fin Field Effect Transistor. It has a 3D fin-shaped silicon channel. The gate surrounds the fin and gives better control. It has low leakage current and low power consumption. FinFET is widely used in modern processors and ICs.

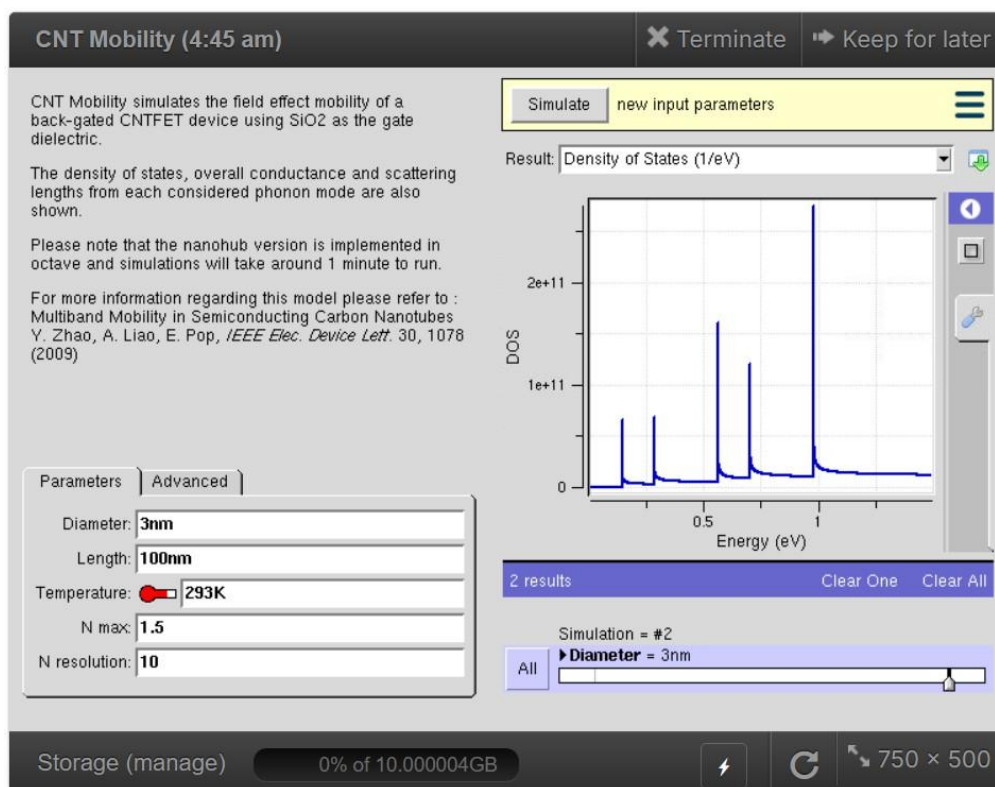
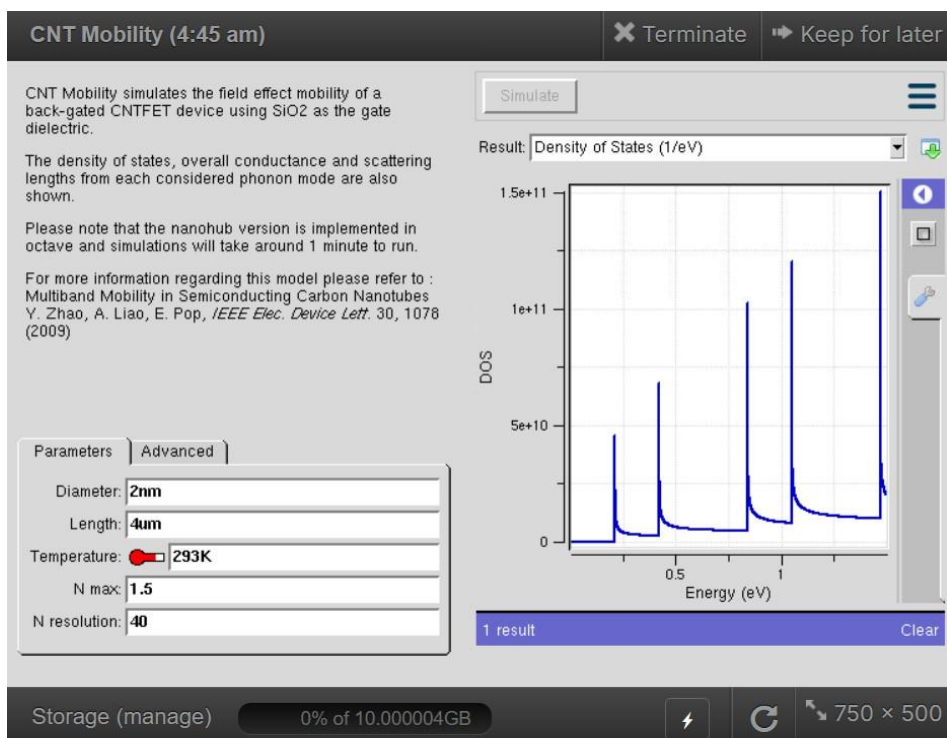


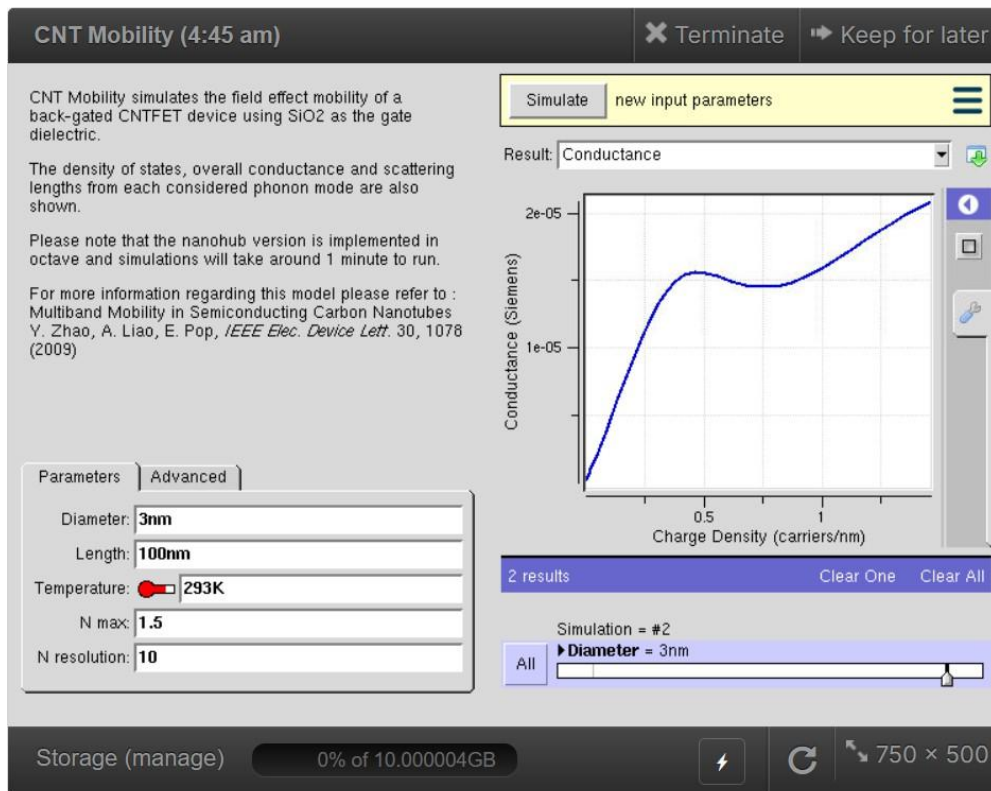


CNTFET

CNFET stands for Carbon Nanotube Field Effect Transistor. It uses carbon nanotubes as the channel material. CNFET offers very high carrier mobility and speed. It consumes very low power during operation. CNFET is used in nanoelectronics and future low-power devices.

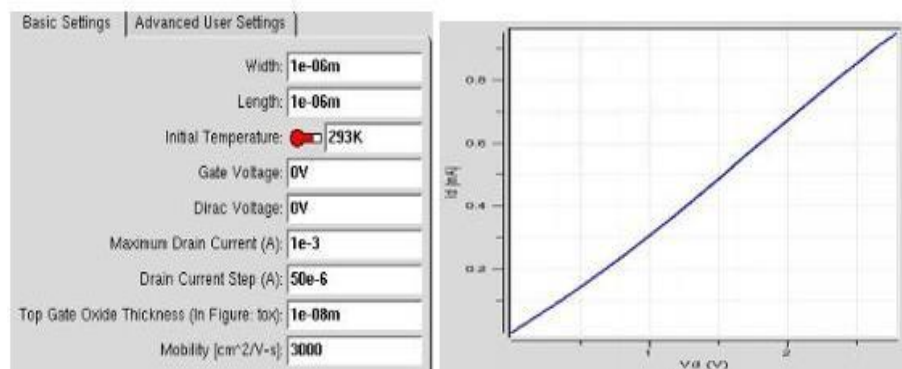


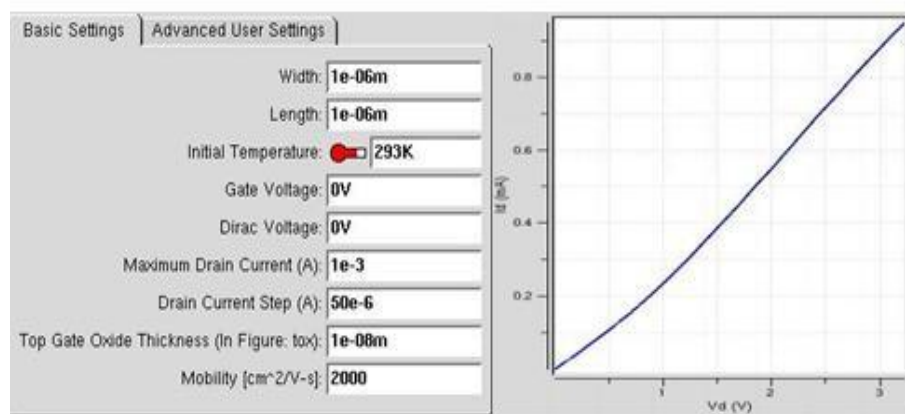
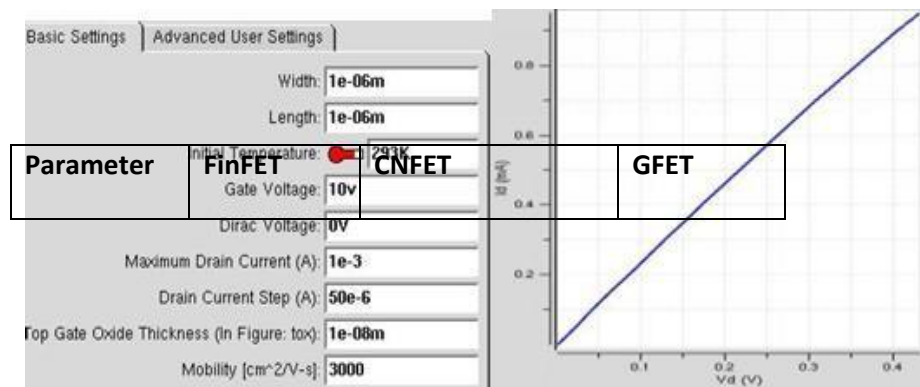




GFET

GFET stands for Graphene Field Effect Transistor. It uses a graphene sheet as the channel material. Graphene has extremely high electron mobility. GFET provides very high-frequency and fast operation. GFET is mainly used in RF and high-speed communication circuits.





Channel	Silicon fin	Carbon nanotube	Graphene
Speed	High	Very High	Very High
Power	Low	Very Low	Low
Leakage	Low	Very Low	Moderate
Application	Processors	Nanoelectronics	RF devices

Conclusion

The characteristics of FinFET, CNFET, and GFET were studied and compared successfully. FinFET provides better gate control, low leakage current, and is widely used in modern processors. CNFET offers very high speed and very low power consumption, making it suitable for future nanoelectronic applications. GFET provides extremely high carrier mobility and is mainly used in high-frequency and RF applications. Hence, each device has its own advantages depending on the application requirement.